# Referee report: *The Lonely Runner Problem on Pair-Sum Lattices* (v4)

Line-level review: flaws, gaps, internal inconsistencies, proposed fixes

# 0. Verdict in one paragraph

The exact part of the paper (Sections 3 to 5.5, Theorem 6.1, Theorem 7.1) is mostly sound, and most of the numbers I recomputed match. The conditional part (Theorem 2.1, Proposition 2.2, Table 1, Section 9) has **three defects that break it as stated**: (i) hypothesis Hq is false as written, because of the paper's own Lemma 9.6; (ii) the depth bound "never exceeds T+3" contradicts the paper's own depth formula; (iii) conclusion (iii), "the cell is empty, hence LRC holds at that rung", does not follow. On top of that, the v2 to v4 revisions were never carried through the whole text. The abstract, Section 1.2, Section 1.3, Section 9.7 and Section 11 still describe the v1 hypothesis set (H0, H2m, "two hypotheses"), and several "now proved" or "closed outright" claims contradict the dependency ledger. Section 9.7 reads like a lab diary, not a final statement. Below, **[C]** = critical (a claim is false or doesn't follow), **[M]** = major (an inconsistency or gap a referee would block on), **[m]** = minor (typo, notation, wording). Every recomputation I did is listed in Section 9.

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# 1. Critical issues

## [C1] Hq is refuted by Lemma 9.6 (the reflection fiber)

Table 1 states Hq “for every admissible cascade history S and next fiber i”, with Gamma “uniform over … dilations”. Lemma 9.6 proves that fiber -j imposes exactly the same constraint as fiber j (the bad sets are symmetric, and U is symmetric under j -> -j). Take a nonempty history S that has already processed fiber j, and let the next fiber be -j. Then |S intersect A| = |S|, and Hq would need Gamma >= p^(m-1)/|Sol| for every S. That is the inverse density (about 256/c1, so above 10^3 at T=8). With that Gamma, r = Gamma*B/p^d >= 1, so Proposition 2.2 never contracts. The same thing already happens at i = 1 for j = -j0. Table 10 shows this directly: small-zone gamma values go up to 2.09 x 10^4, so any Gamma that is uniform over all dilations makes the theorem vacuous.* ***Fix:*** *restate Hq over a fiber set that holds one representative per +/- pair and excludes the resonant zone (lambda-bar <= Lambda0). Then make H2r a statement that enough non-resonant representatives* exist*, rather than adding N_res to the depth (see C4).

## [C2] The fiber budget is |U|/2, not |U|

For the same reason as C1, only about |U|/2 = ((T-2)k + eps - 1)/2 fibers are independent. Theorem 2.1(ii) and Proposition 9.5(d) compare the depth against |U| = (T-2)k. At k = 2 the worst case T+3 is larger than T-2+(eps-1)/2, so the “closure margin at every T >= 8, k >= 2” fails at k = 2 (and at k = 3 for T = 8). It also raises a question about the data: the measured maximum depth of 9 at T = 8, k = 2 (p = 17, |U| = 12) is *more* than the 6 independent fibers. Either the reported depths count reflected duplicates, or the cascade order processes both j and -j. State the convention and recount.

## [C3] “Never exceeds T+3” contradicts the depth formula

The leading term of C(T,k) is log(c1 ov^(m-1)) / [(m-1)log(p/ov) - log(c1 Gamma)]. Its numerator grows like (m-1)log k and its denominator tends to a constant, so the depth grows like log k / log(T/(T-6)) with **no upper bound**. Proposition 9.5(c) admits this (“approaches T-4 only at k >~ (T/(T-6))^(T-4)”), which directly contradicts 9.5(a) “never super-linear, D <= (T-4)+N_res”. Theorem 2.1(i) calls the limit of the *denominator* the “T-flat plateau”, but a denominator is not a depth. Separately, even with Lambda0 = 4, the claimed inequality (T-4)+N_res <= T+3 fails for T >= 9: N_res = 1 + 8(T-2)/T gives 12.2 > 12 at T = 9, 13.4 > 13 at T = 10, and tends to T+5. **Table 9’s own “worst case” column (<= 12.2, <= 13.4) contradicts the “<= T+3” in the abstract, Theorem 2.1 and Proposition 9.5.** **Fix:** state the depth as O_T(log k). That is still far below (T-2)k/2, so (ii) survives for large k. Drop the T+3 claim, or prove a k-uniform bound by some other route.

## [C4] N_res is added, not avoided, and the ledger disagrees with itself about resonant fibers

Proposition 2.2 adds N_res “before the contraction count begins”. But the cascade is free to choose its fiber order, so resonant fibers never need to be processed at all. Meanwhile Hq claims uniformity over *all* dilations (which would make H2r redundant), while H2m is restricted to lambda-bar > Lambda0. Either Hq covers resonant fibers (then N_res is unnecessary, and C1 applies) or it does not (then Hq must say so). Also, “H2m is the i = 1 case of Hq” is false as stated: the dilation ranges differ, and so do the constants (gamma0 vs Gamma).

## [C5] Conclusion (iii) and row EQ: cell emptiness does not imply LRC at the rung

Theorem 2.1(iii) says “the cell (1K, T-3U) at p^2 is empty; and by the lossless reformulation equivalence, LRC holds at that rung”. Table 1’s EQ row says “cell emptiness at the rung implies LRC there”, and Sections 1.3 and 11.1 repeat it (“hence … the conjecture at the covered rungs”). This is a non sequitur. (TAU-n) requires *every* V to have a certifying pair at *its own* pair-sum moduli. Ruling out one cell type at moduli of the form p^2 leaves untouched: prime moduli (where coverings do exist, the rigid zone, unknown at n >= 6), pq and other composites, and all the other kernel/unit cells. This directly contradicts the paper’s own disclaimers (“does not … reduce the conjecture”). Note also that T = 8..15 (n = 7..14) is already settled by [4] to [7], so for those rungs the conditional theorem adds nothing new about LRC. **Fix:** delete “LRC holds at that rung” and rewrite EQ as what it actually is: Corollary 3.3 (the identity M = max tau).

## [C6] Hq is close to circular at late cascade steps

Once |S| * Gamma * |Sol| / p^d < 1, Hq *asserts* S intersect A = empty, which is the conclusion itself. A multiplicative density inequality with no additive error term is false for small structured sets in general (take S inside A). The calibration “depths <= 4 / exit time <= 2” measures exactly the regime where Hq stops being an inequality and becomes the claim. **Fix:** use an inequality with an error term, |S intersect A| <= Gamma|S||A|/p^d + E(k), and redo Proposition 2.2 (contraction until |S| ~ E, then a separate kill argument).

### [C7] H1g's threshold omits Gamma

Proposition 2.2 needs c1*Gamma < (p/ov)^(m-1). H1g only asks c1 <= (T/(T-6))^(m-1) e^(-eta). The sentence "the denominator is uniformly positive under H1g's margin clause" is false unless Gamma <= e^eta. Put Gamma into the margin clause, or bound Gamma separately.

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# 2. Stale-revision inconsistencies (v1/v2 text never updated)

| Location | Says | Should match |
|---|---|---|
| Abstract, last sentence | "the **two** hypotheses of Theorem 2.1 are open" | three (Hq, H1g, H2r) |
| Section 1.2 item 5 | "two open obligations … strand census (H1g) and sharp pair-line constants (**H2m**)" | Hq/H1g/H2r; H2m is now a base case |
| Section 1.3 para 2 | "its two substantive hypotheses (H1g and the H2 pair)" | three |
| Section 2, "Where the derivation lives" | "two open obligations (… H2m)" | same |
| Section 9.7 opening | "one proof away: H1g and the sharp H2m" | contradicts both Hq and the "structural gap, not technical" verdict later in the same section |
| Section 11 item 1 | "The two hypotheses … **H0**'s route … If all three close" | H0 was retired in v2 |

| Location | Says | Should match |
|---|---|---|
| Section 11 item 9 | “Theorem 2.1 closes it under H1g and the H2 pair”; also “whether it survives a third rung is untested” followed by “does survive the third and fourth rungs” | one internally consistent statement |
| Section 11 item 10(a) | “the remaining two lemmas” | “single input S” (Section 6.4) |
| Section 7 intro | “close modulo exactly the **two** lemmas” | Cor. 7.4: “single unbounded input S” |
| Section 9.7 zonotope paragraph | “harmonic m=5 … refuted exactly, rho >= 97/288” | v4: rho = 16/47 exactly (11.10 already lists 97/288 as superseded provenance) |
| Section 9.7 | “Three candidate third routes … none attempted here”, then later “The attempt has since been run” | rewrite as a final-state statement |
| Section 10.4 | “open problem 6 of Section 11” (decay) | it is item **7** |
| Section 10.4 | the kernel-world theorem is “the single open mathematical problem the reformulation now has” | contradicts Theorem 2.1’s three open hypotheses, Input S, and Section 11’s ten items |
| Section 6.4 / Section 11.4 | “the entire programme rests on | true at most for the T = 6 p^2/pq |

| Location | Says | Should match |
|---|---|---|
| | a single unbounded input, Input S” | closures, not “the entire programme” |

**[M] “Proved” vs “rests on Input S”.** Remark 5.10 says the (1K,3U) cell “is now empty at every prime p >= 5”. Section 11.3 says “the p^2 side is closed outright”. Section 6.4 and Corollary 7.4 say the same closure “rests on a single unbounded hypothesis, Input S (verified to p <= 113)”. These cannot all be true. Label the closure **conditional on Input S** everywhere.

**[M] “Post-submission” framing in a v4 monograph.** Remark 5.10, Section 11.3, Section 11.4 and Section 11.9 describe “post-submission” work that is now central (Sections 6 to 9). Fold it into the main narrative.

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# 3. Section-by-section, line-level findings

## Front matter, abstract, Section 1

- **[m]** Section 1.4: “the covering threshold index of **Section ??**” is a broken reference (it should point to Section 4.1).
- **[M]** Section 1.4: “The letter k always denotes a lattice time index”. From Section 5.5 to Section 9, k = floor(p/T) everywhere. Pick a different letter for one of them.
- **[M]** Abstract and Section 1.2 item 3: “complete classifications … at 7, 13, 17, 19, 37 (four bad sets) (Theorems 5.2-5.8)”. At n = 5 only 17, 19 and 37 have theorems (5.5, 5.7, 5.8). 7 and 13 get a one-line parenthetical in Section 5.3 with no statement of the counts 20 and 68. Add Theorems K7’ and K13’ with counts, or soften the claim. The same applies to Section 11.6 “every capable prime <= 37, classified”.
- **[M]** Section 1.1: “Bohman-Bohman-Holzman” is a misattribution. The 6-runner (n = 5 speeds) result is Bohman-

Holzman-Kleitman (2001), and the 7-runner result is Barajas-Serra (2008). Neither appears in the reference list. Add both.
- **[m]** Abstract: “93.4% -> 78.2% over V = 24 -> 56” is fine, but it is “+D3 on the augmented battery”. State that once, explicitly.

## Section 3 (equality theorem)

- **[M] Section 3.5, wrong “tight” anchor.** (1,2,3,5) is listed as a tight set, but M(1,2,3,5) = **1/4** > 1/5 (at t = 1/4 every runner sits at distance 1/4). I recomputed it exactly. (1,3,4,7) = 1/5 and (1,3,4,5,9) = 1/6 do check out as tight. Replace it with (1,2,3,4) or another genuine tight set, and re-check what the verification script asserts for this anchor.
- **[M] Remark 3.7: “collapsing equal speeds, which can only decrease M” is wrong.** Removing duplicate speeds leaves g(t), and therefore M, *unchanged*. What changes is the target 1/(n+1), which gets larger. State the WLOG that way.
- **[m]** Remark 3.7 and Remark 3.8 are verbatim duplicates. Delete 3.8.
- **[m]** Definition 3.1 says N >= 3, but Section 3 allows non-distinct speeds and Lemma 3.6 uses N = 2 (V = (1,1)). Write N >= 2.
- **[m]** Lemma 3.5 proof: “(v_j s’ <= …)” should use index p. The proof’s two-sentence “airtight” remark repeats itself; trim it.
- **[m]** Section 3.5: the novelty paragraph repeats “(two rounds, 56 queries, the arXiv ring enumerated for 2024-2026)” and “the correct prior…” twice in a row.
- **[m]** Section 3.5: “the base rung’s open set (3,5,8,13), whose exact loneliness is M = 5/21”. That is correct, and I reproduced it. Say “open for the battery” to avoid confusion with “open” in the LRC sense.

## Section 4 (covering framework)

- **[M] Lemma 4.2 commentary:** “the pair (1,n) on N = n+1 has all residues invertible” is false unless n+1 is prime (for example n = 5, N = 6: residues 2, 3, 4 are not units). Also, “the n+1 lattice points {0, m, …, (j-1)m}” is j points, not n+1.
- **[M] Lemma 4.5 commentary vs Proposition 4.7:** “every covering-capable composite modulus in range is a lift of a

smaller base modulus", yet the base moduli at n = 4 are listed as {6,7,8,11,13,16}, which includes the composites 6, 8 and 16. Either those three are lifts (then they are not "base"), or the sentence is wrong.
- **[M] Remark 4.8 is wrong as stated.** "Equality exactly at the pentagon configuration (1,2) on Z_5m" fails for m >= 2. I computed max_k min(||k||, ||2k||) = 3, 5, 6 on Z_10, Z_15, Z_20, against N/5 = 2, 3, 4. Equality holds only on Z_5. The law also needs a, b nonzero mod N (a = 0 gives 0).
- **[m]** The n = 4 coincidence claim ("verified for all N <= 80") is actually a one-line consequence of Lemma 4.4: with pm <= 2 at T = 5, 4h+1 <= (4N+1)/5 < N. Prove it instead of citing verification.
- **[m]** Notation: M (loneliness) is reused as a modulus in Lemma 4.5, and m is reused for (N-1)/2, the arc count, the lift index and m*. Rename.

# Section 5 (rigid zone)

- **[C] Remark 5.9, first claim is false.** "Every covering configuration at every rigid prime has four distinct antipodal classes". At N = 7 (only 3 classes) all **20/20** n = 5 coverings repeat a class, and at N = 13, **20 of 68** do. I checked this exhaustively. Restrict the claim to 17, 19 and 37 (where I confirm 0 repeats).
- **[M] Remark 5.9, second claim contradicts itself:** "negation is not a symmetry: B_{-w} = B_w, so the negated tuple is a genuinely different tuple whose classes differ". If B_{-w} = B_w, the classes are identical. What is meant is *exponent* negation, e -> -e (that is, w -> w^{-1}, or reflecting the translate class). Say that. Section 12.2 repeats the confusion.
- **[M] Theorem 5.7:** "8 residue realizations per class" should be per *translate containing 0*: 2 classes x 4 translates x 2^3 = 64, so each class has 32.
- **[M] Conjecture 5.18:** "equivalently: the n = 5 rigid zone is exactly {7,13,17,19,37}" is not equivalent, because the conjecture is about primes and the zone also includes composites. Add "and no composite is unit-capable".
- **[M] Section 5.4 vs Section 7:** "the next micro-cases … 7*17 = 119 … all outside the scanned composite range", yet Section 7 scans

N = 119 (and 115, 143). Update the open problem with what the pq census says about 119.

- **[M] Remark 5.10:** “changes the micro-case picture in three ways”, and then a “Fourth” follows. Also “Lemma A … false at p = 7, 13, where ±-distinct coverings exist and the cells are closed by direct verification”: make it explicit that the Z_p coverings exist but the p^2 cell is still empty. As written it reads as a contradiction.
- **[m]** Section 5.3: “write e(cls(g)^t) = t” should be e(g^t) = t. And 2^m = -1 mod N is necessary but not sufficient for cls(2) to generate C. Say that 2 is a primitive root mod 19 and mod 37.
- **[m]** Theorem 5.2 proof: “b in {2,5}, c in {3,4}” does not match the listed pair (3,5). Phrase it as “the unordered pair hits {2,5} and {3,4}”.
- **[m]** Theorem 5.3: the proof assumes the three starts are distinct (|T| = 3). Add one line: two dominos cover at most 4 < 5.
- **[m]** Table 4: the factorization column is missing for 63, 69 and 75, and “open part of the range (49-111)” leaves out odd composites 9 to 45 (45 has 7 of 10 classes reachable, so it is safe, but say so).
- **[M] Lemma 5.15(a), wrong proof of a true bound.** “gamma’ is at most (n - gamma_max)/(t-1)” is backwards: the largest of the remaining gaps is at *least* their average. Correct one-line proof: any two adjacent gaps sum to n minus the other t-2 gaps, each of which is >= 1, so the sum is <= n-t+2.
- **[m]** Lemma 5.14: write floor(6k^2/(6k+1)). The “for d <= 2k trivially” clause is redundant, since the quadratic inequality already covers it.
- **[m]** Lemma 5.16, case d = 3k-1: at s = 2k+2, |T| = 2 gives span* = 3 + 2 = 5, not 6. This is harmless (G2 = 6 still holds via Gamma2(C)).
- **[m]** Notation in Section 5.5 clashes with the global notation: T (rung vs band set), h (threshold vs run height), sigma, C, M4/M5. See Section 6 below.
- Positive: I independently brute-forced G2(X_d^(s)) <= 2k+1 for every prime p = 1 (mod 6), 31 <= p <= 500, d in [3, 3k-1], both s, and found 0 violations. The “45 primes, 10,410 triples” count matches d in [2, 3k].

# Section 6 (group-ring translation)

- **[M] Theorem 6.1(i) has a two-line algebraic proof.** In R_p, 1_i(z)(1 - z^{d_i}) = z^{a_i}(1 - z^{d_i s_i}) (telescoping), so Q = P * sum_i 1_i = P(Phi_p + E) = PE, because (1 - z^d)Phi_p = 0. The Fourier, Gauss's-lemma argument is unnecessary. Use the short proof and keep the Fourier one for (iii).
- **[M] Count inconsistencies:**
  - Table 5 and Table 14 say "all 15 primes 5 <= p <= 61", but there are 16 primes there, and Table 14's own list has 16 entries.
  - Table 14 says "0 ±-distinct coverings at p >= 11", but Proposition 6.4 says p = 13 has 8 ±-distinct instances.
  - Table 5 says "Lemma A exceptions … all 52", but Section 6.2 says the true solution sets are 272 + 8 = 280, and that "the identity check ran on the full solution set". 52 = 50 + 2 is the *witness* count. Say which set was checked.
- **[M] Table 6 vs Proposition 6.5:** the (2,2) counts are "16 (M1), 48 (M2), 32 (M3)", but Table 6 gives M2 = {2k,2k,2k+2} -> 32 and M3 = {2k,2k+1,2k+1} -> 48. Either the labels or the numbers are swapped.
- **[M] What is actually proved:** Proposition 6.5 is stated "with at most one foot per AP", but the counts are claimed for all configurations, including ov = 2 multisets where two feet are possible. Proposition 6.6 covers ov in {0,1} only, while the Lemma A side has ov = 2 (counts 288 and 48). So the statements "(1,1) family is proved" and "(2,2) family is proved" in Section 6.4 are overclaims. Also "the epsilon = 2 counts" should be "ov = 2" (epsilon is the signed excess).
- **[M] Conjecture 6.7, "Equivalently: (i) … and (ii) …":** this drops the realization and count clause, so it is a consequence, not an equivalence.
- **[m]** q is used for (p-1)/2 in Section 6.3, for p - s1 in the proof of Proposition 6.6, and for the second prime in Section 7. M1 to M5 are kills of Lemma A, multisets in Proposition 6.5, *and* mechanisms in Proposition 8.2. E is both the signed-excess polynomial and a set in Lemma B. Rename them.
- **[m]** "p <= 251 (prior harness)" vs "p <= 250 (prior relaxed census)".

- **[m]** The relationship between the “24 resp. 312 pinned placements” and the “covering counts 28/68” of Input S is asserted (“which is why”) but never explained.

## Section 7 (pq side)

- **[M] Coverage overclaim.** Corollary 7.4 says “verified here for N <= 143”, but the eight moduli {35, 55, 77, 91, 95, 115, 119, 143} leave out **65, 85 and 133**. There are 11 semiprimes pq <= 143 with 5 <= p < q. Section 6.5 and Section 11.4 say “ten semiprime moduli” and Table 14 says “10 moduli” for the (1,1,2) cells. Reconcile all of them and either run the missing three or restate the range.
- **[m]** The proof of Theorem 7.1(ii) writes p = 6m + eps, while the statement uses k. The symbol r is reused.
- **[m]** Table 7 “profile dichotomy (1)-(ii)” is garbled, and “|Y_a^c| … 32 q” lost the fraction 2/3 in typesetting. That happens in several places (5.6, 5.17, 7.1, 7.3); check the PDF rendering of .

## Section 8 (higher rungs)

- **[M] Lemma 8.1:** delta0 = (eps^2-1)/T is not an integer in general (for T = 7, eps = 2 it is 3/7). Define M with a floor, or restrict eps.
- **[M] Lemma 8.1 proof:** “the footprint computation is the one of Section 5.4 verbatim”, but Section 5.4 contains no footprint computation. Point to the companion note, or include it.
- **[M] Proposition 8.2:**
  - “seventeen primes p in {19, …, 137} plus the boundary p = 17”, but the list has **16** entries.
  - “No ±-distinct covering at any k >= 2 (eps in {3,5,7}) … the k = 2 boundary carries twelve ±-distinct configurations” contradicts itself unless those twelve are at eps = 1.
  - “At eps = 5, k = 3 only … gone by k = 7”: “only” and “gone by” disagree. Say “present at k = 3, absent at the next tested eps = 5 prime (k = 7)”.
  - “three classes: 528, 232, 216 -> 2,784” is ambiguous. It only adds up as 528 + 6x232 + 4x216. State the multiplicities.

    - The M1 derivation ({2k+1, (2k+2)^3} sums to p) holds only at eps = 7, yet the counts are claimed for eps in {3,7}. At eps = 3 the window is {2k, 2k+1}.
    - M5: “complementary arc of length 2k+3” must be covered by an interval of size <= 2k+2, which is impossible. Re-check the arithmetic (3s-2 with s = 2k+2).
- **[M] Proposition 8.3(iii):** six counts are given “at k = 2, …, 11” (ten values of k). Also “tail families 120-170 at k <= 4” vs “decays (183 -> 43)” in Section 8.5 and Table 14 (183 is outside 120-170). And the overlap window [k-eps+4, k-eps+8] holds only for eps >= 4.
- **[M] Proposition 8.4(i):** “the pigeonhole |U| > maxproj holds at every k >= 5 except (p = 29, k = 4)”, but k = 4 is not >= 5. More importantly it **fails at p = 43 = 7*6 + 1** (prime): |U| = 30 against a maximum projection of 30 at k = 6, so the inequality is not strict. Also “|U| … 20 -> 60 over the same range (k = 1..11)” is wrong for k < 4.
- **[m]** Proposition 8.4 controls: “all unit **triples**” for a four-unit cell (1K,4U)? Also “20,000 random families at p in {17,19}” (Section 8.5) vs “20K families at four primes” (Table 14).
- **[m]** Section 8.5: “The two-page computation that would discharge Input S” contradicts Section 9’s own rule (“no claim is made about … the amount of work”). Delete it.

# Section 9 (scaling closure)

- **[M] Lemma 9.2 / Table 8:** “(D-1)^(T-4)” should be the **falling factorial** (D-1)_(T-4). The numbers (840 = 7*6*5*4, and so on) are falling factorials. “Saturation identity P = (D-1)^(T-4)” has the same problem.
- **[M] Table 9 caption:** “derived constants (c1 = 1, Lambda0 = 4)” contradicts the free-parameter disclosure (c1 is data-bounded and Lambda0 is data-chosen). Formula values also differ from Theorem 2.1: 8.5 here vs “~9” there.
- **[M] T = 9 miscalibration:** the measured maximum of 11 is above the predicted 9.2. The caption attributes this to “near-tiling contamination of the **T = 10** median” (wrong rung?) and says it is “decidable only by H2m”, but a hypothesis decides nothing. State plainly that the model under-predicts at T = 9.

- **[M] Reproducibility of the calibration:** the cut ratios, medians and maxima of Table 9 “predate the present artifact bundle”. The quantitative support for Theorem 2.1 (Table 1, Hq: “<= 11 across the three calibration rungs”) is therefore not reproducible from [17]. Re-run and commit it, or remove it from the ledger.
- **[M] Table 10:** “Mid zone: lambda-bar > 4 and lambda-bar < p-4” — the second condition is vacuous, since lambda-bar <= (p-1)/2. Also “90th percentiles in 14-422” for the mid zone is impossible when the mid-zone maximum is 40.6. And “survival exactly 1 at lambda-bar = 1” should be lambda = -1 (lambda = +1 is the anchor itself).
- **[M] Theorem 9.9:**
  - The statement uses (m!)^2 order pairs, the proof uses ((m-1)!)^2.
  - The box side is “<= 2m s_max” but the numerator is 4m s_max + 1.
  - The proof uses b_min(lambda, s’_max), the statement b_min(lambda, 2 s_max).
  - Lemma 9.3 is for equal source and target intervals, but here they differ.
  - A unimodular image of a box is a parallelepiped, not a box; you need ||U||_inf.
  - “k-free” is false as stated, because s_max ~ 2k makes the bound k-dependent unless b_min ~ k (Section 9.7 concedes this).
  - H2m’s row in Table 1 quotes yet another form, “(m!)^2 4^m s_max”.
- **[M] “Chain class proved” (Table 1, H1g status) vs Proposition 9.11, “no proved constant yet does” clear 256.** Lemma 9.8 gives |Sol| ≈ ov^(m-1) with no explicit proved constant (its box includes the negative-o stratum). Change the status to “parametrization proved; constant measured”.
- **[M] Section 9.5, p = 29:** “138 of 715 sets, 710 families” is unclear (138 x 24 = 3,312). And “20/138 admissible, consistent with 0.133” compares a *targeted* (biased) sample with a population rate, which is not a valid consistency check.
- **[m]** Proposition 9.11: “near-constant across rungs”. (T/(T-6))^(T-4) tends to e^6 ≈ 403, so it is near-constant only for T in [8,10].

- **[m]** Lemma 9.10: “cuts the family space 5-to-1 in general” should be “at most m-to-1 (orbits can have stabilizers)”.
- **[M] Lemma 9.13 overreach.** The lemma is true but trivial: a deterministic dynamics on single placements is a partial permutation. It closes only *this* automaton, which tracks one anchor along the drift. Natural transfer operators for the strand census (scanning I position by position with local coverage states) **branch**, and can have interior spectral radius. Section 9.5’s own 2x2 example shows exactly this. So “losslessness forbids generated decay”, “bounded by its own fidelity”, and “the toolkit is provably not the right shape” are not supported. The scope caveat (“bijective drift”) appears once, but the abstract and Section 1.2 state the general version. Also, the “third k-invariance theorem” list miscredits the spacing lemma (the k-cancellation is Prop. 9.4, and it is conditional) and Lemma 9.12 (which is not a k-statement).
- **[M] Section 9.7 structure.** It is a chronological log (“since that disclosure”, “has since been run”, “now on record”) with self-superseding paragraphs. Rewrite it as one final-state section: what is proved, what is open, which routes were tried, and the verdict.
- **[m]** Notation in Section 9: D is both (p-1)/2 and the depth (Prop. 9.5) and the difference multiset (Lemma 9.10). Gamma is both Hq’s constant and Gamma2 in Section 5.5. rho is spectral radius, deepest-hole value, and rho_p/rho_q.
- **[m]** Zonotope record: “exact rational arithmetic at 45 digits” is self-contradictory. And n = 6 is benchmarked against 5/14 in one place and 17/50 in another. State delta(n,m) for each row.

## Section 10 (measurement)

- **[M] Section 10.1:** “the distinction between two kinds of flags:” is followed by nothing; the list is missing. The flag numbering skips P3. “D3” is never defined.
- **[M]** “The same battery is the one committed at n = 4, ported verbatim (P4 at {7, 13, 17})”. At n = 4 the rigid primes are {7, 11, 13}, so this is not verbatim. Explain the mapping.
- **[m]** Section 10.2: the n = 4 open set (3,5,8,13), “its only live pairs are (5,13) and (8,13) with effective triples summing to 8 = 3+5 …

covering at N = 16", is garbled. The covering pair is (3,13) at N = 16, and (5,13) and (8,13) are pairs at N = 18 and 21. Rewrite.
- **[m]** "n = 5 sspeeds" (typo). Residual increments recomputed from the table are +3.80, +5.35, +4.08, +2.05 (the paper has 5.36, 4.07, 2.06). Either cite the unrounded data or round consistently.
- **[m]** Mixing set-level percentages with pair-level decompositions (1,798,577 pairs) without saying so.

## Section 11 and Section 12

- See Section 2 above for the stale items.
- **[M] Section 12.4 vs [17]:** "deposited for public access" vs "[17] unpublished artifact bundle". Also "no dependencies beyond CPython 3.12 with numpy", yet pool_sat.c is in the bundle.
- **[M]** "Every number in Sections 10-12 appears verbatim in a committed log" leaves Sections 2-9 uncovered. The artifact list has no scripts for Lemma 9.12, the Lemma 9.13 session, the T = 9 stride samples, or the Input S range p <= 113 (pm_colliding_close.py is listed only to p <= 43 in Table 14). Close the gap or label those numbers "uncommitted".
- **[m]** Table 14: "3,3 M" should be "3.3 M". "classification predicate (126-82,251 per modulus)": n = 4 at N = 7 has only C(8,3) = 56 multisets, so either n = 4 wasn't covered or the range is wrong.
- **[m] References:** [21] the name is Brändén, and the page range is likely 821-891; check it. Check the Davenport-Rado title and the attribution in [18]. Add Bohman-Holzman-Kleitman and Barajas-Serra. [22] needs an arXiv ID. "Companion notes", "Lemmas A/B/T, Corollaries T1-T3" and "Theorems 2-3 of the companion notes" need a citable reference.

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# 4. Proposed improvements (priority order)

1. **Rebuild Theorem 2.1 honestly.** (a) Fiber set = one representative per ± pair, non-resonant only. (b) Hq with an additive error term, quantified over that set. (c) H1g margin

clause that includes Gamma. (d) Depth bound O_T(log k), with the explicit formula. (e) Conclusion: “the frontier cell at p^2 is empty”, full stop. Delete every “hence LRC at the rung”.

2. **Synchronize the hypothesis vocabulary** (abstract, 1.2, 1.3, 2, 9.7, 11.1, 11.9, 11.10): Hq / H1g / H2r open; H2m a measured base case; H0 retired.
3. **One dependency diagram** (a figure) for the T = 6 closures: proved lemmas -> Input S -> p^2 and pq cells. Use “conditional on Input S” consistently, and stop calling things “closed outright”.
4. **Restructure Section 9.7 and Section 11** as final-state text. Move the attack chronology to an appendix.
5. **Fix the false or garbled statements**: Remark 4.8, Remark 5.9 (both claims), Lemma 5.15(a) proof, the (1,2,3,5) anchor, Remark 3.7, Lemma 4.2 commentary, Conjecture 5.18’s “equivalently”, Proposition 8.4’s p = 43 case, Table 10 percentiles, falling factorial notation.
6. **Reconcile all counts** (15 vs 16 primes, 52 vs 280, 8/10/11 semiprimes, 16 vs 17 primes, 120-170 vs 183, Table 6 vs Prop 6.5, 2 vs 4 control primes).
7. **Notation table** (one page): k, m, M, q, D, T, h, sigma, Gamma, rho, E, M1-M5. Every symbol should currently mean at least two things.
8. **Shorten the epistemic prose.** Phrases like “honest”, “the honest boundary” and “structural gap, not technical” occur dozens of times. The ledger makes the point; repetition weakens it.
9. **Put the short algebraic proof of Theorem 6.1(i) first**, and promote K7’/K13’ at n = 5 to theorems with counts.
10. **Commit or drop the uncommitted calibration** (Table 9’s three-rung record).

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# 5. What I recomputed (and it matched)

- M(3,5,8,13) = 5/21; M(1,3,4,7) = 1/5; M(1,3,4,5,9) = 1/6; M(1,2,3,4) = 1/5. **M(1,2,3,5) = 1/4 (mismatch).**

- Rigid-zone counts: n = 4 at 7/11/13 -> 4/12/4 (0 at 17, 19); n = 5 at 7/13/17/19/37 -> 20/68/16/64/32 (0 at 11, 23, 41).
- K7 and K13 (n = 4) covering pair lists exactly as stated.
- Table 4: every row (h, non-unit classes, reachable) matches.
- Multiset counts 165 / 1,140 / 5,985 / 82,251. Lemma 4.4 at T = 6. Example 5.6 bad sets. Gaps of A, B, C0. S = {0,1,4} at N = 19.
- Lemma 5.14 algebra and endpoints; Lemma 5.16 brute force to p = 500 (0 violations); Remark 5.17 endpoint values at p = 31.
- Table 6 row and column sums (1,368; 504/288/288/288); Prop 6.4 B-side 144.
- Table 8: all bounds, saturations and Wilson intervals; T = 9 projected pools.
- Table 11 normalized values, ratios, margins 4.7/12/79 and 95/1,207/3,803; thresholds 256/243/244.1.
- Lemma 9.8: 10,080 = 420 x 24; 120,960 + 2,280 = 123,240. Renormalization examples at p = 19 and p = 29.
- Section 10 percentages, 861-fold, the 48.4% -> 15.1% verified-tier shares.
- The 2x2 matrix [[0, .9], [.9, .4]] has rho ≈ 1.122.